

# Simplifying Ratios - No Decimals/Fractions

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Find the HCF for these numbers, which is the largest number that will divide into both numbers.

Examples

4, 6. = 2      9, 12. = 3

1 3, 6. = 3      2 2, 8. = 2

3 9, 12. = 3      4 5, 20. = 5

5 12, 14. = 2      6 9, 15. = 3

7 6, 12. = 6      8 15, 30. = 15

Now use the HCF to simplify these ratios. Use division.

9 15 : 10  $\frac{5}{\text{HCF}}$  = 3 : 2  
 $\frac{15 \div 5}{8 \div 4} \quad \frac{10 \div 5}{12 \div 4}$

10 8 : 12  $\frac{4}{\text{HCF}}$  = 2 : 3

11 14 : 35  $\frac{7}{\text{HCF}}$  = 2 : 5

12 24 : 18  $\frac{6}{\text{HCF}}$  = 4 : 3

13 10 : 100  $\frac{10}{\text{HCF}}$  = 1 : 10

Find the HCF then use it to simplify these ratios.

14 4 : 8  $\frac{\quad}{\text{HCF}}$  = 1 : 2

15 9 : 6  $\frac{\quad}{\text{HCF}}$  = 3 : 2

16 12 : 20  $\frac{\quad}{\text{HCF}}$  = 3 : 5

17 30 : 15  $\frac{\quad}{\text{HCF}}$  = 2 : 1

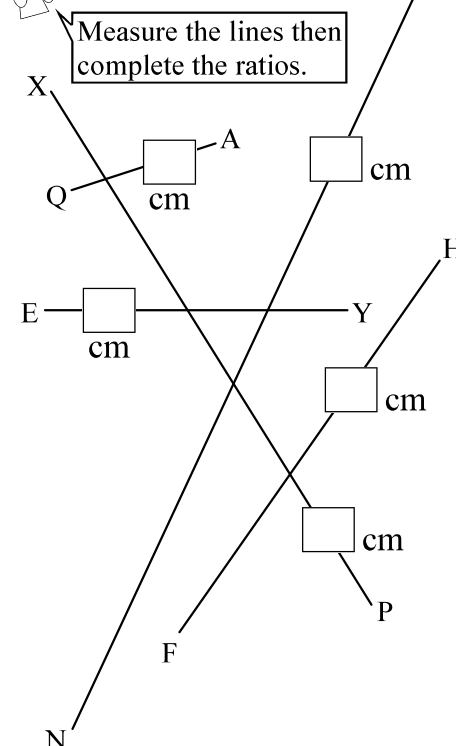
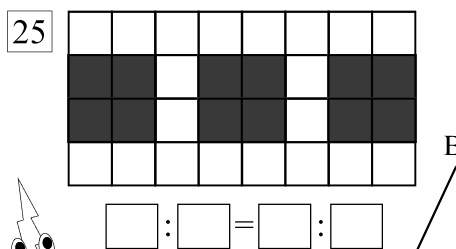
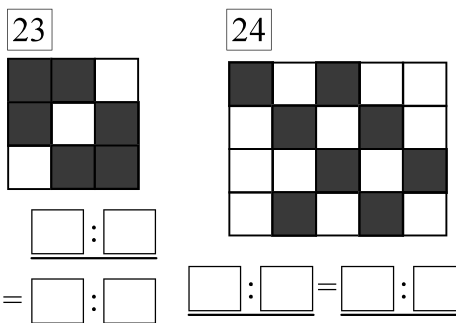
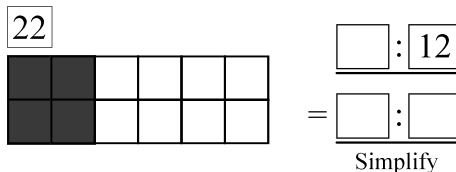
18 12 : 24  $\frac{\quad}{\text{HCF}}$  = 1 : 2

19 100 : 50  $\frac{\quad}{\text{HCF}}$  = 2 : 1

20 18 : 30  $\frac{\quad}{\text{HCF}}$  = 3 : 5

21 24 : 32  $\frac{\quad}{\text{HCF}}$  = 3 : 4

Find the non-simplified ratio for these grids, then simplify. Find the ratio for shaded : total.



26 QA : PX = 1 : 1 = 1 : 1  
 27 NB : HF = 1 : 1 = 1 : 1  
 28 YE : XP = 1 : 1 = 1 : 1  
 29 AQ : BN = 1 : 1 = 1 : 1  
 30 FH : HF = 1 : 1 = 1 : 1

Simplify these, you can use your calculator's fraction key. You have to solve triples mentally.

1st number is smaller than the 2nd number

Examples 8 : 12 = 2 : 3  
 Screen: '2/3'      Calc: 8 [a<sup>b/c</sup>] 12 [=]

A whole number answer is written ':1'

12 : 3 = 4 : 1  
 Screen: '4'      Calc: 12 [a<sup>b/c</sup>] 3 [=]

Change mixed numeral to improper fraction

Screen: '1 1/3'      12 : 9 = 4 : 3  
 Calc: 12 [a<sup>b/c</sup>] 9 [=]  $\frac{\text{SHIFT}}{\text{OR } 2^{\text{nd}} \text{ F}}$  [a<sup>b/c</sup>]

31 16 : 48 = 1 : 3      32 24 : 36 = 2 : 3

33 30 : 6 = 5 : 1      34 60 : 45 = 4 : 3

35 18 : 20 = 9 : 10      36 85 : 50 = 17 : 10

37 36 : 27 = 4 : 3      38 49 : 21 = 7 : 3

39 100 : 95 = 20 : 19      40 50 : 2 = 25 : 1

41 500 : 50 = 10 : 1      42 45 : 18 = 5 : 2

43 4 : 12 : 8 = 1 : 3 : 2

44 9 : 6 : 3 = 3 : 2 : 1

45 25 : 5 : 20 = 5 : 1 : 4

46 16 : 8 : 24 = 2 : 1 : 3

47 10 : 50 : 100 = 1 : 5 : 10

48 20 : 40 : 50 = 2 : 4 : 5

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$$\boxed{48} \quad 20 : 40 : 50 = \boxed{2} : \boxed{4} : \boxed{5}$$